Lab Info

- Traceroute – Section 1.4
- Ping
- ICMP – Section 5.6

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"Real" Internet delays and routes

Traceroute program: measure delay from source to router/final destination along Internet path toward destination

For all routers *i* on the path between Source and Destination:

- Source transmits a set of "probes" to Destination (3 in this example)
- Each set of probes reaches router *i* on path towards Destination
- Router *i* drops the probe and returns a special reply message to sender
- Source measures time interval between probe transmission and reply received
  - Traceroute program reports results of each iteration

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Traceroute output:

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 probes → Three delay measurements from gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms  *
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13 nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16 194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

trans-oceanic link

means no response (probe lost, router not replying)

https://en.wikipedia.org/wiki/Traceroute

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How are probes and replies generated?

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ICMP: Internet Control Message Protocol

- Communicate network-level information
  - Error reporting: unreachable host, network, port, protocol
  - *e.g.,* Echo request/reply
- Network-layer "above" IP:
  - ICMP messages carried in IP datagrams
- ICMP error response message: type, code and first 8 bytes of IP datagram (packet) causing error

Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

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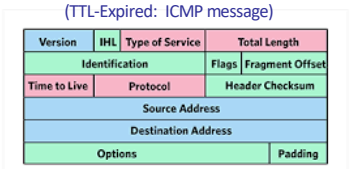
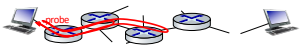
IP Header

(TTL-Expired: ICMP message)





- IP: important information in IP header
  - Destination IP
  - TTL examined and decremented by each router
- How many hops can the probe traverse?

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Traceroute Operation and ICMP

Continual Probing:

- Traceroute Source sends sets of probes with increasing TTL
  - 1st probe set TTL =1; 2nd probe set TTL=2, etc.
- *n*th probe set arrives to *n*th router:
  - Router decrements TTL
  - Router discards probe and sends ICMP message **TTL-Expired** to Source
    - ICMP message includes name of router & IP address
    - When ICMP message arrives back at Source, the RTT is recorded

Note: Sometimes a router doesn't return the message: ***

2 Types of Probes: UDP or ICMP (lab exercise)

When to stop?:

- Probe eventually arrives at Destination host
- Destination returns a final message
- Traceroute stops

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More about the Traceroute Probes

UDP probes

- Probes from source host are carried over UDP to a specific destination port
- At a router, if TTL=0
  - TTL-Expired returned (ICMP msg)
- How to Terminate?
  - At final destination, ICMP Port unreachable returned

ICMP probes

- Probes from source host are ICMP "Echo Request"
- At a router, if TTL=0
  - TTL-Expired returned (ICMP msg)
- How to Terminate?
  - At final destination, ICMP Echo Reply returned

Sometimes there is NO RESPONSE! Oh well ☹

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Ping Program

- Special message (ICMP) sent from a src to dest
- Destination returns a special message (ICMP)
- Time to receive a response (RTT) is recorded
  - Measure RTT
  - Continues until you stop it
  - Special message
    - ICMP Echo Request
  - Special response
    - ICMP Echo Reply
  - Traceroute can also use these ICMP messages for its probes! (or not)

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